

Sarim Shahwar

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Engineering (BEng) in Computer Engineering; GPA: 3.55

Sept. 2021 – Apr. 2026

- **Quantitative & Systems Math:** Probability and Stochastic Processes (MTH 514), Differential Equations and Vector Calculus (MTH 425), Linear Algebra (MTH 141), and Discrete Mathematics (MTH 314).
- **Advanced Coursework (Graduate Level):** Intelligent Systems (EE8209), Distributed and Cloud Computing (EE8229), Computer Vision (CP 8307), and Engineering Algorithms (COE 428).
- **Engineering Economics:** Principles of Engineering Economics (ECN 801) — focused on capital budgeting and risk-adjusted decision making.

EXPERIENCE

Data Intelligence & Algorithm Engineer (Contract)

Apr. 2026 – Jul 2026

LUCA Theory

Montreal, QC (Remote)

- Designed and built the **high-throughput** production backend ecosystem powering the public-facing Hive platform, ensuring real-time delivery of algorithmic insights and predictive intelligence.
- Developed **algorithmic backend** systems using Zod schema validation to enforce strict operational parameters, including multi-variable confidence permutation scoring matrices and **time-locked synchronization**.
- Implemented distributed task execution environments using **Docker** and **GKE**, while syncing high-availability relational states with **Google BigQuery** for real-time identity mapping across parallel **data pipelines**.

Senior Engineer Intern

Jan. 2026 – Apr. 2026

Canada Pension Plan Investment Board (CPPIB)

Toronto, ON

- Refactored and scaled the **Valuations** data system (**Python**, **AWS EKS**, **Athena**, **S3**), powering enterprise valuations, supporting the reporting of **\$700B+** in assets while reducing infrastructure costs.
- Integrated **Pydantic** for **robust data validation** and **schema** enforcement, ensuring high data integrity for complex financial models across **microservices**.
- Developed **scalable data** retrieval patterns using **Amazon Athena** and **S3**, enabling efficient **SQL**-based analysis of large-scale financial datasets.
- Deployed containerized **microservices** and managed Apache **Airflow DAGs** to orchestrate fault-tolerant **ETL pipelines** between upstream finance systems.
- Enhanced release reliability by building **GitHub Actions CI/CD** workflows, automating unit and integration testing to accelerate the team's **development cycle**.

Software QA Intern

Sept. 2025 – Dec. 2025

MEDCAN

Toronto, ON

- Drafted **end-to-end** test cases from PRD/user stories and mapped them to requirements (RTM) covering functionality, performance, reliability, stability, and compatibility with legacy/external systems.
- Built and maintained automation with **Selenium WebDriver** and **TestNG** (POM, data-driven, cross-browser/parallel) and used AccelQ for high-value **regression paths**.
- Executed manual & automated cycles (smoke/regression/UAT); isolated, replicated, and reported defects in Jira with clear evidence; verified fixes and kept TestRail test docs.
- Participated in test plan & case reviews with QA Lead, Dev, and Product; escalated quality risks early and validated deliverables against functional and design specs, **APIs** (Postman) and Database (SQL).

Quantitative Engineer Intern

May 2025 – Aug. 2025

LUCA Theory

Montreal, QC (Remote)

- Built a 'Digital Twin' **ML engine in Python** and various models to simulate market scenarios, achieving **72%** directional accuracy and delivering a realized ROI ranging from **21.4%** to **68.7%** across varying volatility regimes.
- Developed a **RAG-augmented Bayesian learning loop** to dynamically recalibrate asset probability weights based on real-time market feedback, significantly reducing model decay and drawdown without manual intervention.
- Engineered low-latency **pipelines using Selenium** to capture **alternative data** (Alt-Data), integrating granular sentiment signals into a high-performance **BigQuery** backend for real-time alpha generation.
- Orchestrated high-throughput infrastructure via **Docker** and **GKE**, managing auto-scaling compute clusters to execute parallel **Monte Carlo** simulations and visualize trend-line deviations on a custom **JavaScript** dashboard.

PROJECTS

- Limit Order Book Simulator** | *Python, NumPy, SciPy, Pandas* Apr. 2026 – May. 2026
- Built a price-time priority matching engine with $O(\log n)$ insert, $O(1)$ top-of-book, and $O(1)$ cancel, validated by 60 deterministic tests including book invariant assertions.
 - Deployed Avellaneda-Stoikov market-making strategy with inventory-skewed quoting and dynamic spread optimization, achieving positive P&L and bounded inventory across 100K+ event simulations.
 - Calibrated Almgren-Chriss impact model via nonlinear regression on simulated executions and computed optimal liquidation trajectories minimizing cost-variance tradeoff.
 - Parsed Nasdaq ITCH 5.0 binary feed to reconstruct live order books and validated best bid/ask accuracy against official NBBO snapshots.
- AI-Powered Traffic Management System** | *Python, OpenCV, TensorFlow, Vertex AI* Sept. 2025 – Apr. 2026
- Designed and implemented a predictive traffic queue forecasting model using Multiple Linear Regression and Random Forest (Scikit-Learn), integrating municipal traffic data with SUMO simulation outputs.
 - Engineered a full ML pipeline in Python (Pandas, NumPy), including cyclical time encoding, normalization, and window-based feature construction for supervised learning.
 - Integrated Google Cloud Vertex AI to orchestrate final signal-phase decisions based on model outputs and contextual features (weather, congestion thresholds, safety constraints).
 - Optimized model for edge deployment on NVIDIA Jetson Nano using ONNX conversion and post-training int8 quantization (2 - 3x inference speedup)
 - Built a real-time monitoring dashboard using React (Next.js), visualizing delay, throughput, queue length, and model confidence metrics.
- Blockchain System** | *Python, FastAPI, React, PostgreSQL, Docker, Azure* Jan. 2025 – Mar. 2025
- Developed a dynamic blockchain transaction system leveraging FastAPI for high-performance endpoints and MySQL for persistent storage, fronted by React.js.
 - Enforced cryptographic immutability with hashlib, managed efficient JSON handling and validation via Pydantic, and secured endpoints using OAuth, Passlib, and Starlette middleware.
 - Optimized asynchronous data flow by integrating RabbitMQ for message queuing and WebSockets for bi-directional communication, facilitating robust peer-to-peer inter-network exchange.
 - Executed cloud-native deployment strategies by containerizing services with Docker and hosting the infrastructure on Microsoft Azure.
- Basketball Form Analyzer** | *Python, Flask, TensorFlow, PyTorch, Pandas, Git* Jan. 2024 – Apr. 2024
- Developed a real-time basketball shooting analyzer using Python, TensorFlow, PyTorch, Numpy, Pandas, Matplotlib, Scikit-learn, YOLOv8, Mediapipe and OpenCV for object detection, pose estimation, angle calculations, and a result GUI
 - Designed and implemented Flask API's, integrating Machine Learning and Computer Vision to deliver actionable insights, leveraging Deep Learning algorithms to improve shooting accuracy and form with Artificial Intelligence
- GreenGrove Farm Rewards System** | *Python, Django, FastAPI, SQLite, JavaScript, GCP, Git*
- Developed using a modern full-stack environment (Python, Django, FastAPI, SQLite, JavaScript) to build a scalable, secure rewards platform for managing transactions and customer engagement.
 - Built a reward points engine with advanced business logic to support diverse transaction scenarios, tiered point multipliers, and edge-case handling.
 - Integrated a secure authentication and authorization system to protect sensitive user data and transactions.

TECHNICAL SKILLS

Quantitative & ML: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Monte Carlo Simulation, Bayesian Inference, Time-Series Analysis, Stochastic Processes, Reinforced Learning (RL), RAG Architectures.
Languages & Systems: Python, Java, C, SQL, JavaScript/TypeScript, MATLAB; Unix/Linux, Shell Scripting.
Infrastructure & DevOps: Docker, Kubernetes (EKS/GKE), AWS (Athena, S3), Google Cloud Platform (GCP), Azure, Terraform, Apache Airflow, Jenkins CI/CD, GitHub Actions.
Data & Backend: BigQuery, PostgreSQL, MySQL, FastAPI, Flask, Django, RESTful APIs, RabbitMQ, KubeMQ, ETL Pipelines, Databricks.
Developer Tools & QA: Git, Jira, Confluence, Selenium WebDriver, TestNG, Postman, Power BI, Tableau, AccelQ